

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

LABORATORY ASSIGNMENT REPORT

**ASSIGNMENT 2: VARIABLES AND DATA-DRIVEN
AUTOMATION**

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Department	:	CSE (AI & ML)
Course	:	Python for Automation
Academic Session	:	2023-2027
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1. MODULE NAME & EXPERIMENT TITLE

Module Name: Variables and Data-Driven Automation

Experiment Title: Implementation of Variables and Data-Driven Testing in Robot Framework

2. PROBLEM STATEMENT

Develop test cases using Robot Framework that demonstrate:

- Implementation of variables to store data like usernames, passwords, and URLs instead of hardcoding values directly into test steps.
- Creation of a data-driven test that reads test data from an external file and performs the same set of actions across multiple datasets.

3. OBJECTIVE

- To learn variable declaration and management techniques in Robot Framework (`\${SCALAR}`, `&{DICTIONARY}`).
- To decouple test logic from test data for modular code reusability and clean script maintainability.
- To implement Data-Driven Testing (DDT) using Robot Framework test templates and external files (`test\_data.csv`).

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language:** Python 3.11.x
- **Automation Framework & Libraries:** Robot Framework, `SeleniumLibrary`, `DataDriver`
- **IDE & Tools:** Visual Studio Code with Robot Framework extension
- **External Data Format:** CSV (`test\_data.csv`)
- **Concepts:** Parameterization, Data-Driven Templates, Variable Scoping, Test Setup & Teardown, Iterative Assertion.

5. SOURCE CODE / IMPLEMENTATION STEPS

Step 1: External Test Data File (test_data.csv)

test_data.csv - Test Data Sheet

CSV

```
1  ${username},${password},${expected_status}
2  standard_user,secret_sauce,pass
3  locked_out_user,secret_sauce,fail
4  problem_user,secret_sauce,pass
5  invalid_user,wrong_password,fail
```

Step 2: Test Suite Script (assignment2_tests.robot)

6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The code was executed in VS Code terminal using the command `robot assignment2_tests.robot`. The test suite automatically read all 4 dataset rows from `test_data.csv` and executed 4 sequential test iterations.

Screenshot 1: VS Code Integrated Terminal Execution Log

```
PS C:\Users\Ashmita\Automation_Lab> robot assignment2_tests.robot
===== Assignment2 Tests =====
===== Login With External
CSV Data [standard_user, secret_sauce, pass] | PASS | -----
----- Login With External CSV Data [locked_out_user, secret_sauce,
fail] | PASS | -----
----- Login With External CSV Data [problem_user, secret_sauce, pass] | PASS | -----
----- Login With External CSV Data [invalid_user,
wrong_password, fail] | PASS |
===== Assignment2 Tests |
PASS | 4 tests, 4 passed, 0 failed
===== Output:
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html Report:
C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

Screenshot 2: Generated HTML Log Summary File (log.html)

Robot Framework Test Execution Summary: 4 Passed / 0 Failed

Suite Name:	Assignment2 Tests
Data Driver Source:	test_data.csv (4 iterations)
Total Execution Time:	00:00:06.842
Key Assertions Executed:	Input Text, Click Button, Run Keyword If, Page Should Contain Element

7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

Brief Result:

All 4 parameterized test cases generated dynamically from `test_data.csv` passed successfully. The test suite correctly validated positive login flows for valid user credentials and expected error messages for locked out or invalid users.

Observation:

- Using Robot Framework variables (``${URL}``, ``${BROWSER}``, ``&{LOCATORS}``) makes script modification easy; updating a locator in the `*** Variables ***` block automatically propagates across all keywords.
- The `DataDriver` library seamlessly links CSV rows to template parameters without requiring explicit loops or hardcoded repetitive test blocks in `.robot` files.
- Conditional checks using `Run Keyword If` enabled dynamic validation based on expected status (``pass`` vs ``fail``) within a single reusable keyword template.

Conclusion:

In this lab experiment, I successfully implemented variable parameterization and Data-Driven Testing (DDT) using Robot Framework. Decoupling test execution logic from external dataset files (``test_data.csv``) makes automation testing scalable, maintainable, and significantly reduces code duplication when testing modern web applications across multiple data combinations.